

M04 Quick Reference - RC1

Retrieval sheet, not a substitute for practice

DE04-01 Python setup and first program

Run Python from a known environment. | A script is plain text executed by the interpreter.

DE04-02 Variables, data types and operators

Use descriptive variable names. | int, float, str and bool have different behavior.

DE04-03 Strings and formatting

strip removes surrounding whitespace. | lower/title can standardize display cautiously.

DE04-04 Conditions and boolean logic

if/elif/else evaluate conditions in order. | and/or/not combine booleans.

DE04-05 Lists, tuples, sets and dictionaries

Lists are ordered and mutable. | Tuples are ordered and immutable by convention.

DE04-06 Loops and iteration

for iterates over a collection. | while repeats while a condition remains true.

DE04-07 Functions and scope

Parameters are inputs. | return produces an output value.

DE04-08 Modules, imports and project structure

A module is a Python file importable by name. | Imports should not trigger destructive work.

DE04-09 Exceptions and debugging basics

Tracebacks show call stack and failing line. | TypeError, ValueError, KeyError and FileNotFoundError mean different things.

DE04-10 Files, paths, CSV and JSON I/O

Use pathlib for clear paths. | Specify text encoding where relevant.

DE04-11 pandas foundations: load, inspect, filter and aggregate

read_csv loads tabular data. | head/info/dtypes expose structure.

DE04-12 Practical Big-O intuition

O(1) describes constant-time-style access patterns. | O(n) scans grow with input size.

DE04-13 Coding drills: lists and strings

Index/slice carefully. | Prefer readable built-ins when allowed.

DE04-14 Coding drills: dictionaries and sets

Use dict.get for counting patterns. | Use sets for seen/duplicate detection.

DE04-15 Stack, queue, search and sort basics

A stack is LIFO. | A queue is FIFO.

DE04-16 Integrated Python checkpoint and reproducible script habits

Separate read, validate, transform and report functions. | Use explicit input/output paths.